Freight Cost per Kilogram — 2023 vs 2026 vs SeaCoast (all-in basis)

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Rev 2 — corrected: SeaCoast is now shown on an ALL-IN (landed) basis, not air-freight-line-only, so it is compared like-for-like against the 2023/2026 actuals.

1. The two actual shipments

	2023 shipment	2026 shipment (AWB 047-3175-3223)
Origin lane	Coopercabruca → Salvador (SSA)	Black King, Ilheus → Salvador (SSA)
Flight	TAP 047-04136716	TAP 047-3175-3223
Cargo	100 kg cacao nibs	9 articles cacao
Net / gross	100 kg / 104 kg	302.06 kg / 349 kg
Forwarder	Seacoast Logistics (US-side only)	5 Continent #1026-08 (end-to-end)
Status	Delivered	**GATED — awaiting physical arrival**

2. Cost side-by-side (all known cost, USD)

Cost bucket	2023	2026
Forwarder invoice (as billed)	521.03	4,866.33
Brazil-side, billed separately (confirmed)	943.89	136.69
Brazil-side, billed separately (possible, unconfirmed)	0.00	261.75
TOTAL KNOWN	**1,464.92**	**5,003.02 → 5,264.77**

3. Per-kilogram — actuals

Basis	2023	2026 (invoice only)	2026 (+ confirmed extras)	2026 (+ diaria)
Per gross / chargeable kg	**$14.09** (104 kg)	$13.94 (349 kg)	**$14.34**	$15.08
Per net cargo kg	$14.65 (100 kg)	$16.11 (302.06 kg)	$16.56	$17.43

4. SeaCoast estimate — ALL-IN (your "Costing estimates" sheet, SUBTOTAL row)

Weight tier	All-in total (USD)	ALL-IN $/kg	(air-freight line only, for reference)
200 kg	2,361.08	**11.81**	(3.50)
300 kg	2,681.08	**8.94**	(3.40)
500 kg	3,311.08	**6.62**	(3.30)
750 kg	4,136.08	**5.51**	(3.30)
1000 kg	4,911.08	**4.91**	(3.20)
approx 349 kg (interpolated)	approx 2,835	**approx 8.12**	—

The all-in total includes: air freight + export documentation + inland transport (Brazil) + Brazil airport charges + US airline terminal fee + US import handling + US customs clearance + FDA processing + MPF. Cargo value input = $1,000.

5. Corrected comparison — ALL-IN vs ALL-IN

Lane	Weight	ALL-IN $/kg
DAO actual 2023 (Seacoast + Omega)	104 kg	**14.09**
DAO actual 2026 (5 Continent, +confirmed)	349 kg	**14.34**
SeaCoast estimate	200 kg	**11.81**
SeaCoast estimate	349 kg (approx 2026 weight)	**approx 8.12**
SeaCoast estimate	500 kg	**6.62**
SeaCoast estimate	1000 kg	**4.91**

6. What the corrected numbers say

- **All-in, not air-freight-only, changes the conclusion.** SeaCoast's OWN estimate, fully loaded, is approx $8.12/kg at the 2026 cargo weight (349 kg) — versus the DAO's actual $14.34/kg. SeaCoast looks approx 1.8x CHEAPER all-in, not 4-10x on the transport line.
- The gap **widens with volume**: at 1,000 kg SeaCoast is approx $4.91/kg, approx 2.9x below the 2026 actual.
- The DAO's actual per-kg is elevated by **flat per-shipment costs** (customs clearance, terminal, handling, FDA, inland, palletizing) spread over only 104-349 kg. SeaCoast's scale is where its advantage compounds.
- **2026 actual will drift UP, not down:** its invoice is superseded (US-bond line being added back), $414.59 export-customs may be a duplicate, and diaria/reissue are unquantified.

7. Benchmark — express couriers (for completeness, all-in)

Transport basis	Weight band	Indicative USD/kg
DHL Express Worldwide heavy-weight, published retail	68-150 kg	18.70-84.80
DHL Heavy Weight Express service	50-3,000 kg	quote-only, palletized
FedEx vs DHL, same international route	small parcel	FedEx typically 20-40% higher

- Express couriers price on a **parcel curve**, not a freight curve. For 100-350 kg palletized cacao, DHL/FedEx retail all-in lands **$18-85/kg** — roughly **2x-6x the DAO's actual all-in** and **3x-10x SeaCoast's all-in** at comparable weight.
- Express is viable only for urgent, small, high-value parcels — not bulk cacao.
- Caveat: published rates are list/retail; business accounts get up to approx 40% off, heavy tiers are quote-only. A real quote needs exact dims + origin/destination.

8. Caveats

- **SeaCoast SUBTOTAL is a 2023-era quotation**, cargo value pinned at $1,000, bond shown as $0 even with Bond=Yes. At real cacao value, duty/bond adds would lift it — but directionally still well below $14/kg.
- **SeaCoast does not quote below 200 kg** (air-freight minimum) — so the 2023 104-kg shipment could not be priced on this lane.
- **FX (2023):** BRL legs converted at the DAO's standing ledger rate 1 BRL = 0.19405 USD, not the Nov-2023 spot (approx 0.205). At spot, 2023 Brazil-side approx $997 → total approx $1,518 → approx $14.60/kg gross.
- **2026 FX** uses official BACEN PTAX 5.1575 (18/09/2026).
- **Goods value excluded on all sides** — FREIGHT / LOGISTICS only, not landed cost. 2026 goods = $6,473.02 (Rev 15). 2023 goods not ledger-documented.
- **2026 cargo is under the arrival gate** — nothing here touches balances or lands inventory.